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Sugar contents, physicochemical properties, and dental erosive potential of brown sugar boba milk teas, caramel lattes, and oolong teas from popular retail chains in Malaysia

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ABSTRACT

This study characterized the sugar profiles, total dissolved solids, and key physicochemical properties of popular brown sugar boba milk teas (BSBMTs), caramel lattes (CLs), and oolong teas (OTs) from retail chains in Malaysia. High-Performance Liquid Chromatography (HPLC) determined the sugar composition, and results were compared against two handheld refractometers (FNRSI and ATAGO). BSBMTs and CLs, the two dairy-containing categories, exhibited the highest mean total sugar content (8.45 g/100 mL and 6.99 g/100 mL, respectively), largely driven by sucrose and lactose (≥1.58 g/100 mL). In contrast, OTs contained the lowest total sugar (5.89 g/100 mL) and were dominated by fructose and glucose, lacking lactose and maltose. The analysis of physicochemical properties showed that BSBMTs had the highest mean viscosity (91.5 mPa.s), significantly exceeding CLs and OTs. OTs were distinctly acidic (mean pH 5.5), while BSBMTs and CLs clustered near neutral (mean pH ≈6.7). Crucially, the FNRSI meter (mean difference: 3.44 g/100 mL) and the ATAGO meter (mean difference: 5.01 g/100 mL) both showed significant systematic and proportional biases when estimating total sugar content compared to HPLC ($\beta > 0.98$; $p < 0.001$). These findings provide detailed compositional data for popular retail beverages and highlight substantial inaccuracies when non-HPLC methods are used for commercial sugar quantification.

1. Introduction

The global consumption of beverages has seen a significant shift towards sugar-sweetened options, driven by convenience, variety, and effective marketing strategies. This trend is particularly evident in Southeast Asian countries, including Malaysia, where a vibrant café culture and the proliferation of popular retail chains have led to an increased intake of various sweetened drinks (Mohd Hanim et al., 2021). Indeed, the latest Malaysia Health and Morbidity Survey (NHMS) 2024 indicates that a significant proportion of the Malaysian population, particularly adolescents and young adults, frequently consumes prepared and ready-to-drink beverages (6.2 and 7.3 servings/week and 6.5 and 4.6 servings/week, respectively), often exceeding recommended daily limits for free sugar intake (Institute for Public Health (IPH), 2025). Among these, brown sugar boba milk teas (BSBMTs), caramel lattes (CLs), and different types of teas, such as oolong teas (OTs), have gained immense popularity, becoming staple choices for many consumers. While these beverages offer refreshment and indulgence, their potential impact on public health, particularly oral health, warrants close scrutiny. High consumption of sugar-sweetened beverages (SSBs) is a well-established risk factor for both dental caries and dental erosion, two prevalent oral diseases that can lead to irreversible tooth damage and significant health burdens (World Health Organization (WHO), 2015). Dental caries, commonly known as tooth decay, is a multifactorial disease initiated by the fermentation of dietary carbohydrates by oral microorganisms, leading to acid production and subsequent demineralization of tooth enamel and dentin (Rathee and Sapra, 2025). Dental erosion, on the other hand, involves the chemical dissolution of tooth hard tissues by acids not necessarily of bacterial origin, often from extrinsic sources such as acidic foods and beverages (Abou Neel et al.,

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2016). The interplay of various physicochemical properties—including sugar content, pH, titratable acidity, and viscosity—determines the overall cariogenic and erosive potential of a beverage (Aykut-Yetkiner et al., 2013; Fernández et al., 2022; Owens, 2007).

Despite the widespread consumption of BSBMTs, CLs, and OTs in Malaysia, a comprehensive understanding of their specific sugar contents, physicochemical properties, and direct dental erosive and cariogenic potential remains limited in the scientific literature. Consumers often lack awareness regarding the precise sugar levels and acidic nature of these popular drinks, making it challenging for them to make informed choices for their oral health. Furthermore, existing data on beverage properties often relies on general categories, without specific analysis of the unique formulations offered by popular retail chains. The complex composition of these beverages, encompassing various types of sugars (e.g., fructose, sucrose, lactose (Min et al., 2017; Zubairi et al., 2023), varying pH levels (Reddy et al., 2016), and different viscosities, necessitates a detailed investigation to accurately assess their risks. Without such data, public health campaigns and dental professionals face challenges in providing targeted recommendations for mitigating the oral health consequences associated with their consumption.

Therefore, this study aims to bridge this critical knowledge gap by providing empirical data on the sugar content, physicochemical characteristics, and dental erosive and cariogenic potential of commonly sold BSBMTs, CLs, and OTs in the Malaysian market. The findings will serve as a vital resource for public health bodies, dental professionals, and consumers, enabling the development of evidence-based dietary guidelines and awareness campaigns. By highlighting specific brands or beverage types that pose higher risks, this research can facilitate targeted interventions and encourage manufacturers to reformulate their products or provide clearer nutritional information. Ultimately, this study contributes to a more informed approach to oral health promotion in the context of evolving dietary trends in Malaysia and potentially other regions with similar beverage consumption patterns. To achieve this, the objectives of this study were: 1. To determine the total sugar content and individual sugar profiles (fructose, glucose, galactose, sucrose, maltose, and lactose) of selected BSBMTs, CLs, and OTs from popular retail chains in Malaysia using High-Performance Liquid Chromatography (HPLC); 2. To analyse the physicochemical properties, including %Brix, %Total Dissolved Solids (TDS), initial pH, conductivity, and titratable acidity, of these beverages; 3. To assess their dental erosive and cariogenic potential by calculating the Relative Caries Potential Index (RCPI) and evaluating their buffering capacity and viscosity; and 4. To compare these properties and potentials among the different beverage categories and brands. Based on these objectives and the expected composition of these popular retail beverages, we formulate two primary hypotheses: 1. BSBMTs and CLs will exhibit significantly higher total sugar concentration and greater associated cariogenic potential (as measured by RCPI and viscosity) compared to OTs, and 2. Handheld refractometers will show a systematic positive bias, significantly overestimating total sugar content when compared to the gold-standard HPLC method.

2. Methodology

2.1. Study design and sample collection

This study employed an *in vitro* design to analyse the sugar content, physicochemical properties, and dental erosive potential of various beverages. The sampling methodology was designed to capture a representative selection of popular, commercially available, freshly prepared beverages across leading retail chains in Malaysia, specifically focusing on three categories: BSBMTs, CLs, and OTs. A total of 20 BSBMTs, 26 CLs, and 10 OTs popular beverage samples were collected from leading retail chains across Malaysia (Supplementary Table 1) between June to July 2025. The samples were explicitly chosen based on their market popularity, as confirmed by proprietors and consumers. This ensures the samples reflect the beverages most frequently consumed by the Malaysian public, increasing the ecological validity of the findings. Only beverages readily available across all popular retail chains in Malaysia were included. This avoids the inclusion of niche or limited-time offerings, reinforcing the focus on commonly consumed products. The beverages included both high- and low-cost brands across different market segments (budget, mainstream, and premium), which often correlate with variations in ingredient quality, formulation, and sugar content. All beverages were purchased in their standard serving sizes, with regular sugar and ice levels. BSBMT is prepared by first cooking tapioca pearls/boba and infusing them with thick brown sugar syrup, “painting” the inside of a cup with the syrup to create the signature "tiger stripes" effect, and finally adding cooked boba, ice, milk, and chilled black tea together (Science Buddies and De Brabandere, 2019). A caramel latte is prepared by combining a shot or two of espresso with caramel syrup, then topping it with cold milk and ice, often finished with a caramel drizzle and whipped cream. OT is produced by gradual dehydration and moderate bruising of *Camellia sinensis* (L.) O. Kuntze, that clarifies and intensifies its distinctive fruity and floral flavour and fragrance (Zeng et al., 2020). The brewing process involves heating water, adding the tea leaves, performing a quick rinse, and then steeping for varying durations to reveal the evolving flavours with each infusion. OTs sold at popular retail chains in Malaysia are sugar syrup-sweetened and iced by default. The beverages were transported to the laboratory under controlled conditions to minimize changes in their physicochemical properties. The retail price of each beverage type varies greatly. BSBMTs ranged from as affordable as MYR 5 (~USD 1.2) from BMX to as costly as MYR 19.90 (~USD 4.7) from BHT. As for CLs, affordable options available from vending machines or coffee trucks were often priced below MYR 10 (~USD 2.4), while specialty shops could charge up to MYR 20.65 (~USD 4.9), like from a popular USA brand. OTs were relatively more budget-friendly than BSBMT and CL, with most options being under MYR 15 (~USD 3.5). Samples were stored at 4°C and analysed within 24 h of purchase. For each brand and beverage type, two independent samples were collected to ensure reproducibility and statistical validity. For analysis, beverages were left at room temperature, and once the ice had completely melted, each component was separated and weighed for analysis. The brown sugar boba pearls were blended together with the milk tea for further analysis to mimic the actual consumption of BSBMT in humans.

2.2. Total sugar determination

2.2.1. Brix meter

Two Brix meters: FNIRSI JBM-20 Digital Brix Refractometer (FNIRSI, China) costing ~USD 30 and ATAGO PAL-1 Digital Pocket Refractometer (Code 3810; Atago, Japan) costing ~USD 272 were used to measure the sugar content of each beverage in %Brix (total soluble solids, primarily sugars), and their %TDS (total dissolved solids) were calculated from the %Brix using the following conversion formula: $0.85 \times$ %Brix. Both instruments were calibrated according to the manufacturer’s instructions before use, using 10 %, 20 %, 30 %, and 50 % w/v sucrose solutions [Sigma-Aldrich (St. Louis, MO, USA)]. For each sample, three replicates were performed, and the average values were recorded.

2.2.2. Sugar analysis by high-performance liquid chromatography (HPLC)

The individual sugar profiles (fructose, glucose, galactose, sucrose, maltose, and lactose) and total sugar content of the beverages were quantified using High-Performance Liquid Chromatography (HPLC) at an ISO/IEC 17025:2017 and Laboratory Accreditation Scheme of Malaysia-certified laboratory (Permulab Sdn. Bhd., Petaling Jaya, Malaysia) using an in-house protocol ref. no. FH08 (based on HPLC). According to the laboratory, all monosaccharide and disaccharide standards - fructose, glucose, galactose, sucrose, maltose, and lactose - were purchased from Sigma-Aldrich (St. Louis, MO, USA) with confirmed purities of $\geq$ 99.5 %. All standards were prepared in ultrapure

water to create stock solutions and subsequently diluted to generate a series of working standards for calibration curves. HPLC-grade acetonitrile and ultrapure water (resistivity ≥18.2 MΩ·cm) were used for all mobile phase preparations. The concentration of each sugar was expressed as g/100 mL. HPLC was validated for linearity, limit of detection (LOD), limit of quantification (LOQ), accuracy, and precision. Linearity was assessed by analysing standards at seven concentration levels, all yielding correlation coefficients (R^2) greater than 0.999 (detailed calibration curve equations are provided in Supplementary Table 2). LOD and LOQ were calculated based on a signal-to-noise ratio of 3:1 and 10:1, respectively. Only 7 BSBMTs, 10 CLs, and 3 OTs were sent for analysis.

2.3. Physicochemical analysis

The following physicochemical parameters were measured for each beverage sample:

2.3.1. pH and conductivity analysis

The pH and conductivity (mV) of beverages were measured using a calibrated digital pH meter (Eutech pH2700 Precision Benchtop pH and Conductivity Meter, Eutech Instruments, Singapore). Before each measurement, the pH meter was calibrated using standard buffer solutions (pH 4.0, 7.0, and 10.0) at 25°C. The probe was thoroughly rinsed with deionized water and blotted dry between samples. For each sample, three independent measurements were taken, and the average pH and conductivity values were recorded.

2.3.2. Titratable acidity

The titratable acidity, indicative of the buffering capacity, was determined by titrating 10 mL of each beverage sample with 1 M sodium hydroxide (NaOH) solution dropwise in 100 μL increments until a stable pH of 5.5 (demineralisation), pH 7.0 (neutral), and finally pH 10 was reached (Jung and Jun, 2021). The volume of NaOH (μL) required to reach the endpoint was recorded. Each titration was performed in triplicate, and the mean volume was used for analysis.

2.3.3. Dental erosion potential

The dental erosion potential was evaluated following the International Organization for Standardization (ISO) Method which utilizes calcium phosphate ($CaPO_4$) to imitate hydroxyapatite, the primary mineral component found in tooth enamel (Jung and Jun, 2021). The $CaPO_4$ solution was prepared fresh daily and used on the same day. For each measurement, 25 mL of CaPO4 was stirred, and its initial pH was measured. Subsequently, 0.25 mL of the sample was added to the solution, and its pH was measured. The change in pH was measured by calculating the difference in pH before and after the addition of the sample. This process was repeated three times, and the average pH change was calculated.

2.3.4. Viscosity

The viscosity of the beverages was measured using a viscometer (Anton Paar Rheometer MCR92, Graz, Austria) at a constant temperature of 25°C. The shear rate was set at 50 s^{-1} and the temperature was set at 25°C. Three measurements were taken for each sample, and the average viscosity in milliPascal-seconds (mPa.s) was recorded.

2.3.5. Relative caries potential index (RCPI)

The Relative Caries Potential Index (RCPI) was calculated for each beverage to provide a composite measure of its cariogenic potential. The RCPI takes into account total sugar contents and viscosity, which are all known to influence the caries process. The specific formula used for RCPI calculation was: total sugar contents (by gold-standard HPLC) × viscosity (Lee et al., 2014). A higher RCPI value indicates a greater potential for inducing dental caries.

2.4. Statistical analysis

All statistical analyses were performed using GraphPad Prism version 9.0 (GraphPad Software, San Diego, CA, USA)]. Descriptive statistics (mean ± standard deviation) were calculated for all measured parameters. Bland-Altman plots were generated to assess the agreement between different sugar measurement methods (%Brix from FNRSI vs. ATAGO, and total sugars from FNRSI/ATAGO vs. HPLC). The level of significance was set at $p < 0.05$.

3. Results

3.1. Sugar contents and physicochemical properties of beverages

Fig. 1 illustrates the average %Brix and %TDS for various beverage brands across BSBMTs, CLs, and OTs, as measured using FNRSI and ATAGO meters. Fig. 1A shows that for BSBMTs, the FNRSI Average % Brix ranged from 11.0 % (BCP) to 17.9 % (BAJ). The ATAGO Average % Brix values largely corresponded, with BAJ consistently showing the highest sugar content. The FNRSI Average %TDS generally mirrored the %Brix values, indicating that dissolved solids primarily consisted of sugars. For CLs (Fig. 1B), the FNRSI Average %Brix varied, with CBB displaying the highest value at approximately 11.8 %. Brands like CKP and CYY also exhibited relatively high %Brix, while CHWC and CZUS generally had lower %Brix and %TDS. The ATAGO Average %Brix largely corresponded with the FNRSI readings for %Brix, consistently affirming the perceived soluble solid content. In stark contrast, OTs (Fig. 1C) presented significantly lower %Brix and %TDS values. The FNRSI Average %Brix for OTs ranged from approximately 1.6 % (OGC) to 7.1 % (OKOI). The ATAGO Average %Brix again showed consistency with the FNRSI readings, confirming the lower sugar concentration in this beverage type. OCP demonstrated the lowest %Brix and %TDS across all brands and beverage types analysed, whereas OKOI had the highest values within the OT group.

Fig. 2 illustrates the mean concentration of total sugars and breakdown of different sugar types in the beverages, as measured by gold-standard HPLC. For BSBMTs (Fig. 2A), sucrose was consistently the most abundant sugar, contributing significantly to the total sugar content of each bar. BTL displayed the highest total sugar content among BSBMTs (summing to 8.69 g/100 mL), primarily due to its substantial sucrose concentration of approximately 7.11 g/100 mL. Fructose and glucose were also notably present (BGC being highest for both: 1.32 and 1.65 g/100 mL, respectively), along with isolated occurrences of maltose and lactose in BCB and BTL, respectively. Overall, BSBMTs showed high total sugar levels, reflecting the significant contribution from sucrose and lactose. CLs (Fig. 2B) also showed sucrose as the dominant sugar and a major contributor to their total sugar content. CBB recorded the highest sucrose concentration among CLs, at 7.09 g/100 mL, contributing to its higher total sugar content. Significant levels of lactose were observed across several CL brands (CSB being the highest at 2.58 g/100 mL), consistent with their dairy content, adding to the overall sugar load. Fructose and glucose were present in smaller concentrations. Total sugar content varied among CLs, with some brands notably higher than others. OTs (Fig. 2C) displayed considerably lower total sugar concentrations compared to both BSBMTs and CLs. In contrast also, sucrose dominated OKOI only (2.48 g/100 mL), but was absent from OGC and OBT. The overall total sugar content in OTs was markedly lower, and lactose and maltose were not detected in these beverages. Fig. 2D provides an aggregated view of the mean sugar distribution by beverage type. BSBMT (“B”) demonstrated the highest overall mean total sugar content (8.45 g/100 mL), predominantly from sucrose (4.18 g/100 mL) and lactose (1.58 g/100 mL), followed by roughly equal amounts of fructose (1.06 g/100 mL) and glucose (1.08 g/100 mL). CLs ("C") followed, with a mean total sugar content of 6.99 g/100 mL, with significant contributions from sucrose (2.15 g/100 mL) and lactose (1.76 g/100 mL). OTs ("O") exhibited the lowest

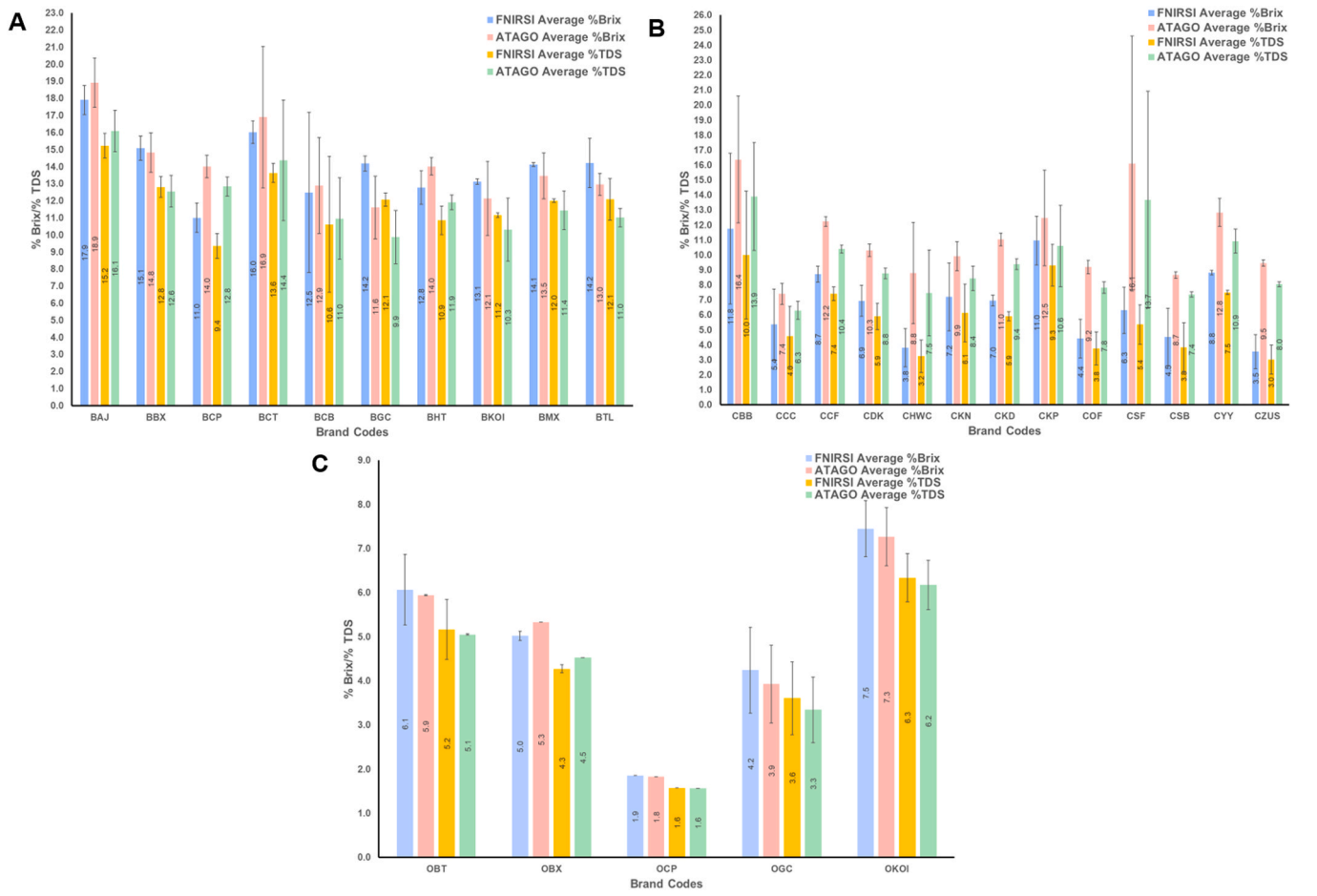


Fig. 1. Average %Brix and %TDS of Beverages. (A) Average %Brix and %TDS of brown sugar boba milk teas (BSBMTs); (B) Average %Brix and %TDS of caramel lattes (CLs); (C) Average %Brix and %TDS of oolong teas (OTs). Data are presented as mean ± standard deviation. %Brix was measured by FNRSI and ATAGO meters, and %TDS was measured by a FNRSI meter.

overall mean total sugar content (5.89 g/100 mL), composed mainly of sucrose (2.48 g/100 mL), fructose (1.89 g/100 mL), and glucose (1.52 g/100 mL), with no detectable lactose or maltose in the aggregated data for this category. Galactose was consistently absent across all beverages and brands.

Fig. 3 presents Bland-Altman plots evaluating the agreement and potential biases between different methods utilized for measuring sugar content: the FNRSI and ATAGO refractometers, and High-Performance Liquid Chromatography (HPLC). Fig. 3A illustrates the comparison of %Brix measurements between the FNRSI and ATAGO meters. A mean difference of −2.04 % (95 % CI: −8.03 % to 3.96 %) was observed, indicating that, on average, the FNRSI meter reported a %Brix value 2.04 % lower than the ATAGO meter. A one-sample *t*-test ($p < 0.001$) confirmed a significant systematic bias between these two handheld instruments. However, linear regression analysis (B = 0.048, $p = 0.650$) suggested an absence of proportional bias. Fig. 3B compares the total sugar content (g/100 mL) obtained from the FNRSI meter against that from HPLC. The mean difference was 3.44 g/100 mL (95 % CI: −3.02–9.90 g/100 mL), signifying that the FNRSI meter generally yielded higher total sugar concentrations compared to HPLC. A significant systematic bias was confirmed by a one-sample *t*-test ($p < 0.001$). Furthermore, linear regression analysis (B = 1.09, $p < 0.001$) indicated a significant proportional bias, implying that the discrepancy between the FNRSI meter and HPLC increased with higher mean total sugar concentrations. Similarly, Fig. 3C presents the comparison of total sugar content (g/100 mL) between the ATAGO meter and HPLC. A mean difference of 5.01 g/100 mL (95 % CI: −0.40–10.42 g/100 mL) was found, suggesting that the ATAGO meter also systematically overestimated total sugar content compared to HPLC, with a larger mean difference than the FNRSI meter. A one-sample *t*-test ($p < 0.001$) confirmed a significant systematic bias, and a significant proportional bias was also observed via linear regression (B = 0.982, $p < 0.001$), indicating that the overestimation by the ATAGO meter increased with higher sugar concentrations. Collectively, Fig. 3 demonstrates significant systematic biases and, for total sugar content comparisons with HPLC, proportional biases when using handheld refractometers (FNRSI and ATAGO) for sugar quantification. Both handheld devices tended to overestimate total sugar content compared to the HPLC reference method, with the magnitude of overestimation increasing at higher sugar concentrations.

3.2. Dental erosive potential

The mean initial pH and conductivity of the beverages are shown in Fig. 4. The mean initial pH values for BSBMTs generally clustered around neutrality, ranging from 6.5 (BHT) to 7.1 (BCT), while the mean conductivity values for BSBMTs varied considerably, ranging from 6.6 mV (BCT) to 34.1 mV (BHT), indicating differences in their ionic composition (Fig. 4A). No consistent inverse correlation between pH and conductivity was observed across all BSBMT brands. For CLs (Fig. 4B), the mean initial pH values ranged from around 6.4 (CYY and CKD) to 6.7 (CSB, CBB, CSF), with most brands maintaining a pH close to neutral. The mean conductivity values for CLs were generally lower than those observed for BSBMTs, mostly falling between 3.5 mV (CZUS) and 43.0 mV (CDK). CYY notably displayed an exceptionally low mean

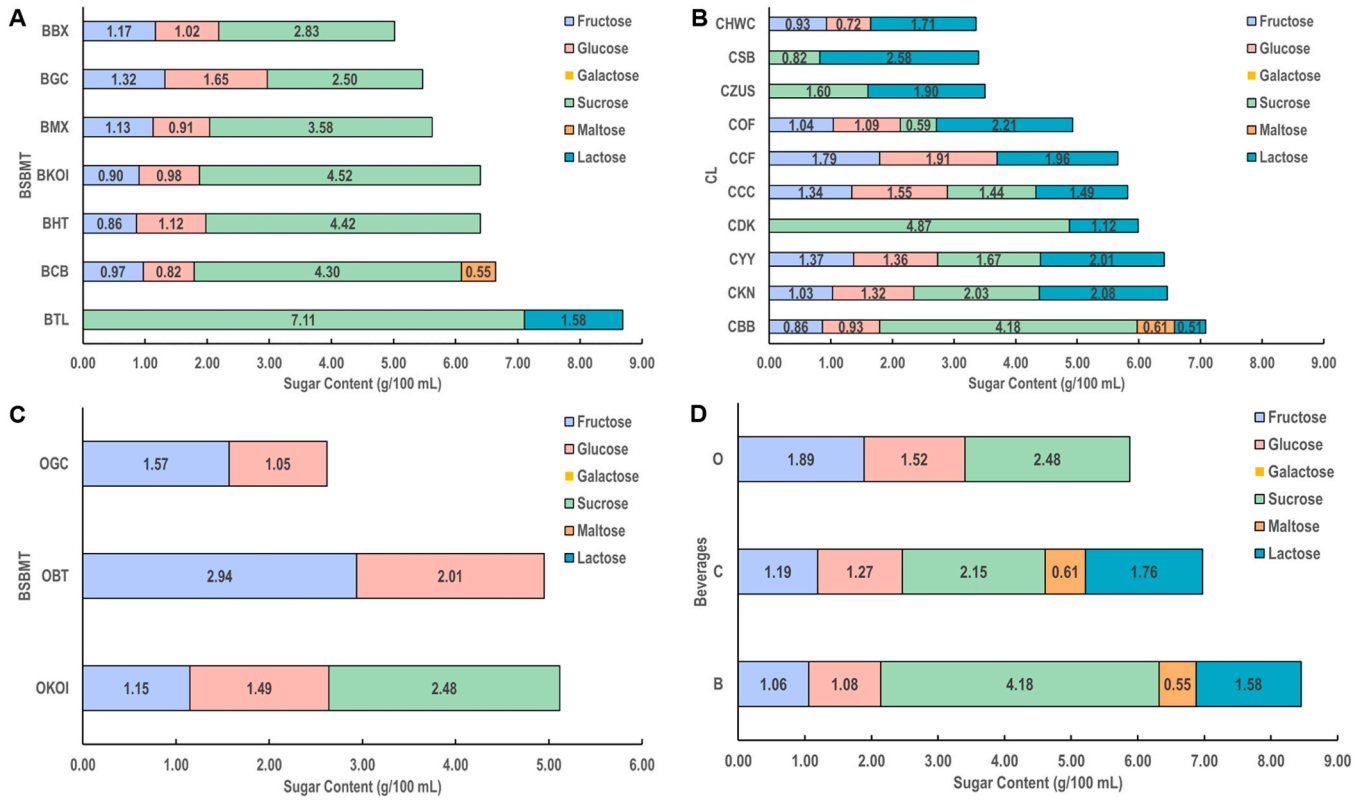


Fig. 2. Mean Sugar Concentrations of Beverages by High-Performance Liquid Chromatography (HPLC). (A) Mean concentration (g/100 mL) of different sugars in BSBMTs; (B) Mean concentration (g/100 mL) of different sugars in CLs; (C) Mean concentration (g/100 mL) of different sugars in OTs; (D) Mean concentration (g/100 mL) of different sugars by beverage type (B: Brown Sugar Boba Milk Tea, C: Caramel Latte, O: Oolong Tea). Sugar concentrations were measured by high-performance liquid chromatography (HPLC).

conductivity value, approaching 0 mV. OTs (Fig. 4C) consistently exhibited lower mean initial pH values compared to both BSBMTs and CLs, signifying a more acidic nature, ranging from 5.3 (OGC) to 5.9 (OBT). Conversely, the mean conductivity values were considerably higher, ranging from 81.0 mV (OBT) to 98.0 mV (OGC), suggesting a higher concentration of dissociated ions in OTs despite their lower pH. Fig. 4D provides an aggregated summary of the mean initial pH and conductivity for the three beverage types. It clearly shows that BSBMTs (B) and CL (C) have similar mean initial pH values, both around 6.7–6.8, indicating they are near neutral. Their mean conductivity values are relatively low, with CL slightly lower than BSBMT. In sharp contrast, OT (O) exhibit a distinctly lower mean initial pH, averaging around 5.5, confirming their acidic nature. Concurrently, OTs display a significantly higher mean conductivity, approaching 80 mV, suggesting a greater concentration of ionized substances compared to the other two beverage categories.

The titratable acidity of the beverages, measured by the pH change after adding NaOH, is presented in Fig. 5. Fig. 5A summarizes the mean initial pH and titratable acidity by beverage type, with BSBMTs and CLs having higher initial pH and OTs having a lower initial pH. Although OTs have the lowest starting pH, they required the least mean volume of NaOH of approximately 100 μL to reach pH 10. On the other hand, CLs required the highest mean volume of NaOH of around 261.54 μL to reach pH 10 checkpoint. These results would suggest that OTs, while acidic in nature, do not resist pH change while caramel lattes can withstand change and this could contribute to their dental erosion potential. The potential of the beverage samples to cause dental erosion was tested by calculating the pH change of the $CaPO_4$ solution after addition of samples. Fig. 5B shows that all BSBMTs showed a positive pH change, indicating an increase in pH upon titration. Brands such as BAJ, BBX, BGC, and BKOI had a pH change greater than 0.05, suggesting higher buffering capacity. For CLs (Fig. 5C), the pH change was variable, with some brands showing a negative change (i.e., CZUS, CSF, CHWC, CDK), and others showing a positive change. The OTs (Fig. 5D) generally showed a negative pH change, with OKOI having the largest negative change (-0.02). Although the changes were minimal, these results demonstrate that certain CLs and OTs can reduce pH, increasing acidity and thus, may contribute to dental erosion, with CDK, CSF, and OKOI have higher potentials to cause erosion compared to other brands.

Fig. 6 shows the relative caries potential index (RCPI) and viscosity of the beverages. The RCPI, calculated based on total sugar content and viscosity, provides an overall measure of cariogenic potential. Fig. 6A shows that BSBMTs had high RCPI values, with BMX having the highest (514.5). The high viscosity values (mPa.s) also contributed to this high RCPI, with values ranging from 2.8 to 91.5 mPa.s. CLs also had high RCPI values, with the highest for CKN (25.9; Fig. 6B). In contrast with BSBMTs, the viscosity for CLs only ranged from 1.8 to 4.0 mPa.s. OTs had very low RCPI and viscosity, with values below 7 and around 1.2, respectively (Fig. 6C). The overall comparison by beverage type (Fig. 6D) shows that BSBMTs (B) had the highest RCPI and viscosity, followed by CLs (C), and finally OTs (O) which had the lowest values for both parameters.

4. Discussion

This study provides a comprehensive analysis of the sugar content, physicochemical properties, and dental erosive potential of popular BSBMTs, CLs, and OTs from retail chains in Malaysia. Our findings highlight significant variations in these parameters across different beverage types and brands, with important implications for dental health, particularly in a region with a high consumption rate of such drinks.

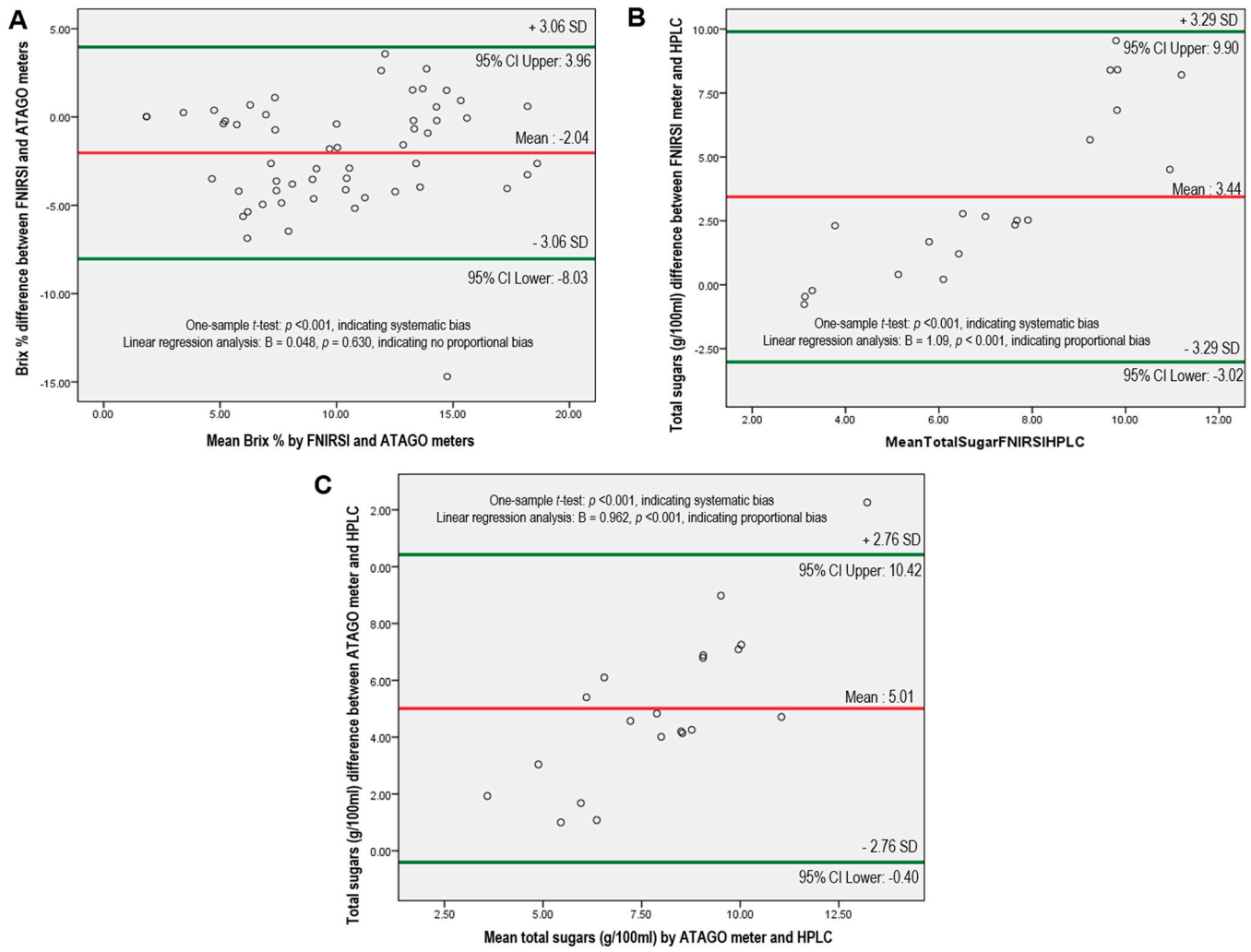


Fig. 3. Bland-Altman Plots Comparing Sugar Measurement Methods. (A) Bland-Altman plot comparing the difference in %Brix measurements between the FNRSI and ATAGO meters against the mean %Brix of both meters; (B) Bland-Altman plot comparing the difference in total sugar (g/100 mL) between the FNRSI meter and HPLC against the mean total sugar of both methods; (C) Bland-Altman plot comparing the difference in total sugar (g/100 mL) between the ATAGO meter and HPLC against the mean total sugar of both methods.

Brix meters and HPLC were used to determine the sugar concentration of the samples. Two types of Brix meters were used: one affordable (FNIRSI) and the other was professional grade (ATAGO). %TDS was calculated from %Brix to give an estimate of the soluble solids present in the beverages, and this correlates with the quality of the beverages, especially in coffee (Várady et al., 2022). It is often used to determine the coffee strength, and the ideal TDS range according to the Specialty Coffee Association of America guidelines is 1.15–1.35 %, depending on the type of coffee. Most of our samples exceeded this range, though the ideal range can vary based on beverage type. Our analysis of sugar measurement methods revealed a notable discrepancy between the quick, handheld refractometers and the more precise HPLC method. The handheld devices consistently showed a systematic overestimation of sugar content compared to the gold standard HPLC, suggesting that while they offer a convenient estimation, their use for accurate nutritional labelling or precise research requires caution. This highlights the importance of using robust analytical methods to accurately assess the true sugar content of complex food and beverage matrices.

Our results confirm that BSBMTs and CLs are highly concentrated in sugars, with fructose and sucrose being the predominant types (Min et al., 2017). The total sugar concentrations in these beverages are alarming – with BSBMT having an average total sugar content of 8.45 g/100, similar to a previous study (Min et al., 2017) – and some brands containing over 11 g of sugar per 100 mL. This level of sugar intake is a major public health concern, as it far exceeds the WHO recommendation for daily free sugar consumption (World Health Organization (WHO), 2015). They also exceeded the threshold of 5 g of total sugars per 100 mL for the Malaysian sugar tax, but remain untaxed under the current sugar tax policy (Sritharan et al., 2024). Although often used interchangeably, dental caries and erosions have different causative agents. Dental caries or decay are caused by acid-producing bacteria, especially *Streptococcus mutans* (Rathee and Sapra, 2025), while dental erosions do not involve microorganisms but rather occur when external chemicals from food, beverages, and drugs cause the mechanical wear and tear of teeth enamel (Jager et al., 2012). Sugars have been found to promote the growth of acid-producing bacteria within the oral cavity by disrupting the symbiotic balance of the microbiome (Giacaman et al., 2018; Pitts et al., 2021). This is because these bacteria metabolize dietary sugars and produce acids, lowering the pH within the cavity. The resulting acidic environment promotes enamel demineralization, increasing susceptibility to dental caries. This finding aligns with extensive international studies that consistently identify SSBs as major contributors to increased risk of dental caries and erosion (Valenzuela et al., 2021). The problem is significantly exacerbated by global trends showing that the prevalence of high SSB consumption among young adults aged 15–39 years increased by almost two-fold

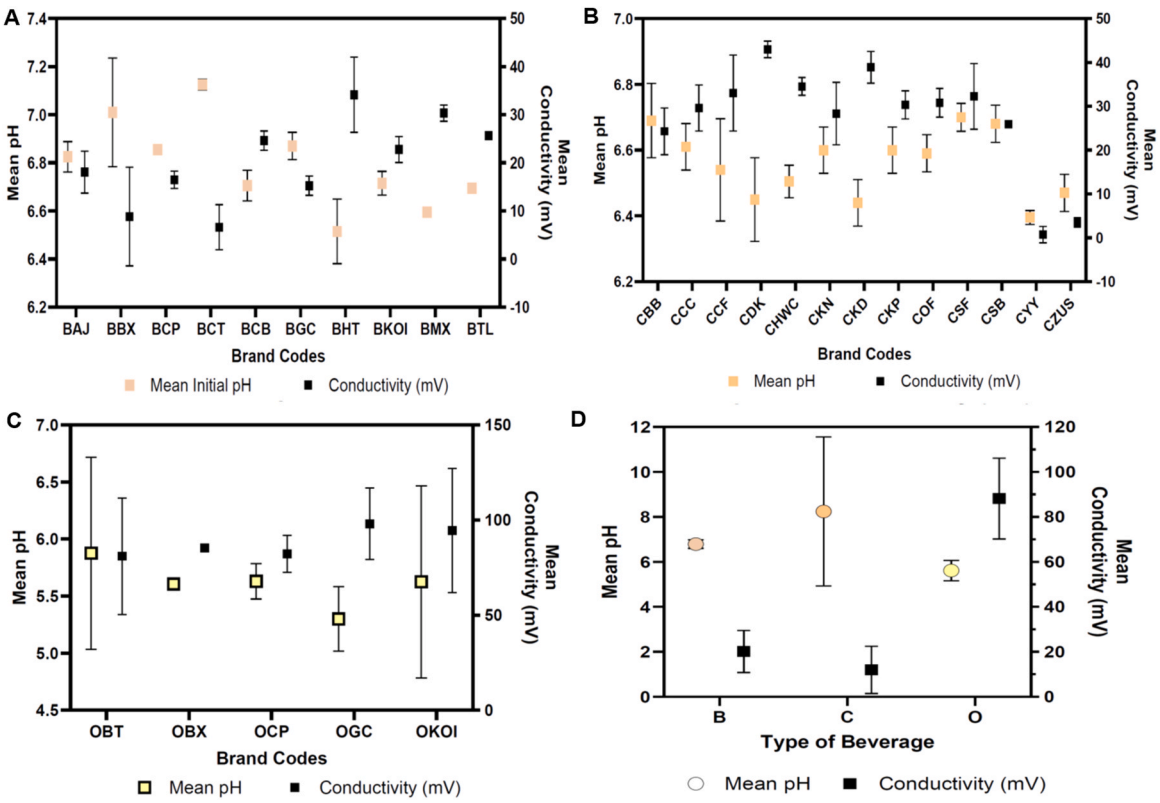


Fig. 4. Mean Initial pH and Conductivity of Beverages. (A) Mean initial pH and conductivity (mV) of BSBMTs; (B) Mean initial pH and conductivity (mV) of CLs; (C) Mean initial pH and conductivity (mV) of OTs; (D) Mean initial pH and conductivity (mV) by beverage type (B: Brown Sugar Boba Milk Tea, C: Caramel Latte, O: Oolong Tea). Data are presented as mean ± standard deviation.

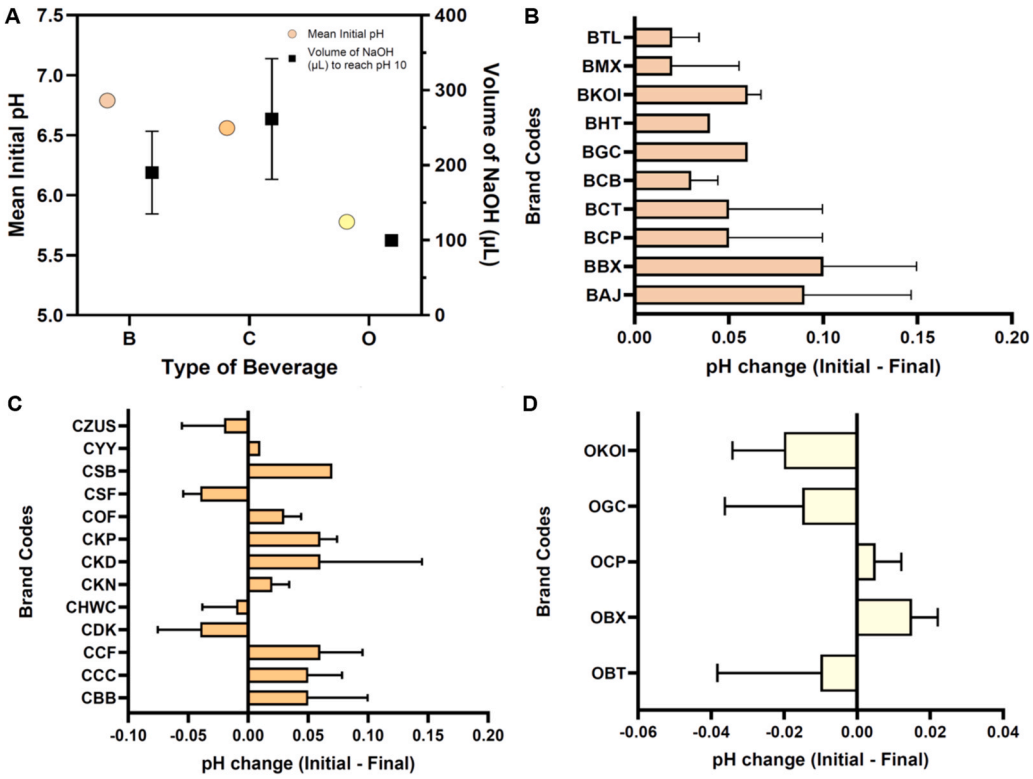


Fig. 5. Mean Initial pH and Titratable Acidity of Beverages. (A) Mean initial pH and volume of NaOH (μL) to reach pH 10 by beverage type (B: Brown Sugar Boba Milk Tea, C: Caramel Latte, O: Oolong Tea); (B) pH change (Initial - Final) of BSBMTs; (C) pH change (Initial - Final) of CLs; (D) pH change (Initial - Final) of OTs. Data are presented as mean ± standard deviation.

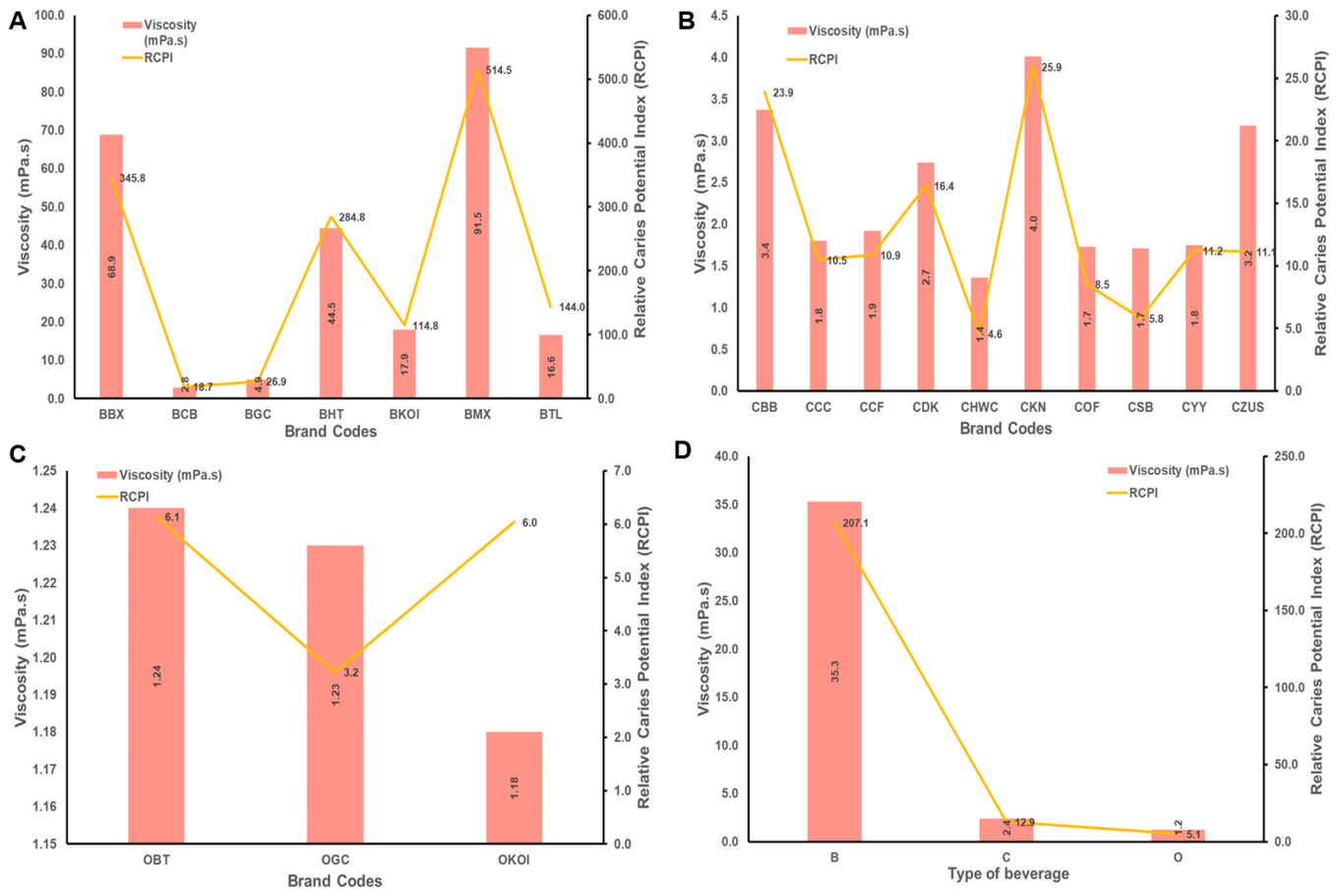


Fig. 6. Viscosity and Relative Caries Potential Index (RCPI) of Beverages. (A) Viscosity (mPa.s) and RCPI of BSBMTs; (B) Viscosity (mPa.s) and RCPI of CLs; (C) Viscosity (mPa.s) and RCPI of OTs; (D) Viscosity (mPa.s) and RCPI by beverage type (B: Brown Sugar Boba Milk Tea, C: Caramel Latte, O: Oolong Tea). Viscosity and RCPI were measured using a viscometer and calculated, respectively.

from 6.58 % in 1990–11.13 % in 2021 (Ge et al., 2025).

The physicochemical properties of the beverages also play a crucial role in their potential for dental damage. The pH of a drink is a primary indicator of its erosive potential, with values below the critical pH of 5.5 for enamel dissolution posing a significant risk (Harper et al., 2021). While most of the BSBMTs and CLs in our study had an initial pH above this critical threshold, their high sugar content and viscosity still present a considerable cariogenic risk. Furthermore, titratable acidity, which measures the total amount of acid and buffering capacity, provides a more complete picture of a beverage's erosive potential than initial pH alone. The titratable acidity results showed a varying buffering capacity among the brands, with some BSBMTs and CLs demonstrating a greater ability to resist pH changes, potentially slowing down demineralization. In stark contrast, the OTs exhibited a lower initial pH, putting them in the acidic category. While their low sugar content suggests a lower cariogenic risk, their inherent acidity could still contribute to dental erosion with frequent consumption. This aligns with existing literature on acidic beverages, which are known to cause irreversible loss of tooth structure regardless of their sugar content (Jameel et al., 2024; Mathew et al., 2018a; Rajeev et al., 2020).

The high viscosity observed in the BSBMTs and CLs is a critical finding with significant implications for oral health. While some studies suggest that high viscosity might reduce the rate of enamel erosion by slowing down the diffusion of acids (Aykut-Yetkiner et al., 2014; Sakae et al., 2018), its primary oral health consequence in these sugary drinks is the prolonged adherence of fermentable carbohydrates to the tooth surface - significantly slowing oral clearance rates - a mechanism widely recognized in international dental literature (Atkinson et al., 2021). This sustained contact provides a long-lasting energy source for cariogenic bacteria, thereby extending the duration of the acid challenge and overwhelming the natural salivary buffering mechanisms. This combination of high sugar content and high viscosity is reflected in the exceptionally high RCPI values calculated for BSBMTs and CLs. The sticky, high-sugar nature of the brown sugar syrup and the presence of tapioca pearls further exacerbate this problem by trapping sugars in and around teeth, a factor consistent with other research on the oral health impact of bubble tea (Zubairi et al., 2023). OTs, with their low viscosity and low sugar content, showed a commensurately low RCPI, positioning them as a less harmful option for dental health.

In short, the distinct dental risks posed by different beverage types are categorized into two primary threat mechanisms: dental erosion and dental caries (cariogenic risk). Traditional packaged ready-to-drink acidic beverages, such as sodas, juices, and energy drinks, are the main cause of dental erosion due to their low pH and high titratable acidity (Maladkar et al., 2022; Mathew et al., 2018b), a risk also noted for OTs, but not for BSBMTs and CLs, which have a less acidic pH (> 5.5). Conversely, BSBMTs and CLs are classified as high-viscosity, high-sugar beverages and consequently pose a major cariogenic risk. The combination of high sugar content and high viscosity in these drinks significantly slows down oral clearance, thereby prolonging the contact time of fermentable carbohydrates on the teeth. This extended exposure overwhelmingly promotes bacterial acid production, as reflected in their exceptionally high RCPI values, whereas traditional soft drinks and OTs present a lower cariogenic risk due to their lower viscosity and/or sugar content, which allows for quicker oral clearance.

A key strength of this study is its comprehensive approach, which went beyond simple sugar measurements to include a full panel of physicochemical properties and a composite caries risk index. The use of

HPLC as a reference method for sugar analysis is also a significant strength, providing a more accurate assessment and revealing the limitations of using handheld meters for this purpose. Furthermore, this is one of the few studies to specifically investigate the dental health implications of these popular beverages within the Malaysian market, besides a previous one (Zubairi et al., 2023), making the findings highly relevant to the local context and public health policy. However, this study also has several limitations. This study's findings are limited by its cross-sectional, *in vitro* design, which fails to account for crucial physiological factors like salivary flow, buffering capacity, and remineralization processes. Furthermore, the generalizability is restricted as the analysis did not quantify the independent effects of common add-ins (e.g., tapioca pearls, cheese foam) (Min et al., 2017), or the impact of retailers' reduced-sugar options. Consequently, future research must prioritize *in vivo* and longitudinal clinical trials to generate real-world correlation data necessary to establish a direct link between consumption and dental outcomes. This data will be vital for developing educational campaigns that pivot towards providing consumers with granular, evidence-based recommendations for customizing their beverage choices and understanding personalized risk factors.

4.1. Policy implications

These findings support NHMS 2024's concern regarding the excessive SSB consumption while highlighting the loophole in Malaysia's current sugar tax policy. The tax applies to packaged SSBs that contain more than 5 g of total sugars for every 100 mL of beverage but excludes freshly prepared SSBs, even though they often exceed this threshold. A revamp of the tax policy to include such beverages could increase public awareness regarding their consumption of hidden sugar content in popular beverages. This could also encourage businesses to reduce the use of sugar in the preparation of their products. Alternatively, promoting the usage of sugar substitutes can be used to create healthier beverages. For example, Stevia is a natural plant-derived sweetener, particularly from the *Stevia rebaudiana* plant (Ashwell, 2015; Siraj et al., 2019). Stevia is zero-calorie and has an intense sweetness. It has even been found to lower blood sugar levels (Siraj et al., 2019), which are attributes that play to its appeal, especially among patients who are managing their diabetes and weight (Zare et al., 2024). It has also been discovered to be nonfermentable and impair bacterial survival, making it beneficial for dental health (Siraj et al., 2019). Although the long-term effects of Stevia are still mostly under researched, a previous study found that Stevia could increase pro-inflammatory cytokines and urea levels (Farid et al., 2020).

5. Conclusions

In conclusion, this study confirms that popular, freshly prepared BSBMTs and CLs in Malaysia pose a significant dental health risk—a finding that mirrors the global crisis of excessive SSB consumption. The extremely high sugar content and viscosity in BSBMTs and CLs drive elevated RCPI values, necessitating immediate and comprehensive action. Policymakers must swiftly revamp the national sugar tax to include these high-risk, point-of-sale beverages and mandate gold-standard HPLC testing for accurate labelling. Concurrently, manufacturers must prioritize product reformulation for significant sugar reduction and substitution with non-cariogenic alternatives like Stevia to mitigate documented public health harm, while public health efforts must emphasize both the hidden sugar dangers in these indulgent drinks and the erosive risk associated with frequent consumption of acidic beverages like oolong teas.

CRediT authorship contribution statement

Victoria Olubunmi Olarewaju: Supervision, Data curation. **Kai-Wei Liao:** Supervision, Methodology. **Megan Kee Ern:** Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Yee-How Say:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT-5 and Gemini 2.5 Flash in order to improve the readability and language of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of Competing Interest

The authors confirm that no financial interests or personal connections exist that could have influenced the outcomes or interpretation of the research presented in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jfca.2025.108600.

Data availability

The datasets used and analysed during the current study are available from the corresponding author on reasonable requests.

References

Abou Neel, E.A., Aljabo, A., Strange, A., Ibrahim, S., Coathup, M., Young, A.M., Bozec, L., Mudera, V., 2016. Demineralization-remineralization dynamics in teeth and bone. Int J. Nanomed. 11, 4743–4763. https://doi.org/10.2147/IJN.S107624.

Ashwell, M., 2015. Stevia, nature's zero-calorie sustainable sweetener: a new player in the fight against obesity. Nutr. Today 50, 129–134. https://doi.org/10.1097/NT.0000000000000094.

Atkinson, F.S., Khan, J.H., Brand-Miller, J.C., Eberhard, J., 2021. The impact of carbohydrate quality on dental plaque pH: does the glycemic index of starchy foods matter for dental health? Nutrients 13, 2711. https://doi.org/10.3390/nu13082711.

Aykut-Yetkiner, A., Wiegand, A., Bollhalder, A., Becker, K., Attin, T., 2013. Effect of acidic solution viscosity on enamel erosion. J. Dent. Res 92, 289–294. https://doi.org/10.1177/0022034512473115.

Aykut-Yetkiner, A., Wiegand, A., Ronay, V., Attin, R., Becker, K., Attin, T., 2014. In vitro evaluation of the erosive potential of viscosity-modified soft acidic drinks on enamel. Clin. Oral. Invest 18, 769–773. https://doi.org/10.1007/s00784-013-1037-9.

Farid, A., Hesham, M., El-Dewak, M., Amin, A., 2020. The hidden hazardous effects of stevia and sucralose consumption in male and female albino mice in comparison to sucrose. Saudi Pharm. J. 28, 1290–1300. https://doi.org/10.1016/j.jsps.2020.08.019.

Fernández, C.E., Brandao, A.C.S., Bícego-Pereira, E.C., Del Bel Cury, A.A., Cury, J.A., Tenuta, L.M.A., 2022. Effect of pH and titratable acidity on enamel and dentine erosion. Clin. Oral. Invest 26, 5867–5873. https://doi.org/10.1007/s00784-022-04544-4.

Ge, C., Xiong, J., Zhu, R., Hong, Z., He, Y., 2025. Global burden of high sugar-sweetened beverage consumption among young adults. Diabetol. Metab. Syndr. 17, 259. https://doi.org/10.1186/s13098-025-01845-y.

Giacaman, R.A., Pailahual, V., Díaz-Garrido, N., 2018. Cariogenicity induced by commercial carbonated beverages in an experimental biofilm-caries model. Eur. J. Dent. 12, 027–035. https://doi.org/10.4103/ejd.ejd_188_17.

Harper, R.A., Shelton, R.M., James, J.D., Salvati, E., Besnard, C., Korsunsky, A.M., Landini, G., 2021. Acid-induced demineralisation of human enamel as a function of time and pH observed using X-ray and polarised light imaging. Acta Biomater. 120, 240–248. https://doi.org/10.1016/j.actbio.2020.04.045.

IPH), 2025. National Health and Morbidity Survey (NHMS) 2024: Nutrition - Key Findings.
Jager, D.H.J., Vieira, A.M., Ruben, J.L., Huysmans, M.C.D.N.J.M., 2012. Estimated erosive potential depends on exposure time. J. Dent. 40, 1103–1108. https://doi.org/10.1016/j.jdent.2012.09.004.
Jameel, R.A., Zaidi, S.J.A., Siddiqui, S., Rehman, A., Gul, J., Saquib, M., Rahim, Z.A., 2024. The effects of beverage erosion on enamel: evaluating surface characteristics and loss of calcium and phosphate ions. Discov. Appl. Sci. 6, 439. https://doi.org/10.1007/s42452-024-06153-0.
Jung, E.-H., Jun, M.-K., 2021. Evaluation of the erosive and cariogenic potential of over-the-counter pediatric liquid analgesics and antipyretics. Children 8, 611. https://doi.org/10.3390/children8070611.
Lee, K.S., Kim, N.-J., Lee, E.-H., Cho, J.-W., 2014. Cariogenic potential index of fruits according to their viscosity and sugar content. Int J. Clin. Prev. Dent. 10, 255–258. https://doi.org/10.15236/ijcpd.2014.10.4.255.
Maladkar, S.R., Yadav, P., Muniraja, A.N.A., Uchil, G.S., George, L.V., Augustine, D., Rao, R.S., Patil, S., Sowmya, S.V., Haragannavar, V.C., 2022. Erosive effect of acidic beverages and dietary preservatives on extracted human teeth—an in vitro analysis. Eur. J. Dent. 16, 919–929. https://doi.org/10.1055/s-0041-1742131.
Mathew, S., Luke, A., Walia, T., Masri, A., Jamal, H., Pawar, A., 2018a. Effect of fruit juices and other beverages on loss of tooth structure. Pesqui. Bras. odontopediatria Cl. íN. Integr. 18, 1–9. https://doi.org/10.4034/PBOCI.2018.181.22.
Mathew, S., Luke, A., Walia, T., Masri, A., Jamal, H., Pawar, A., 2018b. Effect of fruit juices and other beverages on loss of tooth structure. Pesqui. Bras. odontopediatria Cl. íN. Integr. 18, 1–9. https://doi.org/10.4034/PBOCI.2018.181.22.
Min, J.E., Green, D.B., Kim, L., 2017. Calories and sugars in boba milk tea: implications for obesity risk in Asian Pacific Islanders. Food Sci. Nutr. 5, 38–45. https://doi.org/10.1002/fsn3.362.
Mohd Hanim, M.F., Md Sabri, B.A., Yusof, N., 2021. Online news coverage of the sugar-sweetened beverages tax in Malaysia: content analysis. JMIR Public Health Surveill. 7, e24523. https://doi.org/10.2196/24523.
Owens, B.M., 2007. The potential effects of pH and buffering capacity on dental erosion. Gen. Dent. 55, 527–531.
Pitts, N.B., Twetman, S., Fisher, J., Marsh, P.D., 2021. Understanding dental caries as a non-communicable disease. Br. Dent. J. 231, 749–753. https://doi.org/10.1038/s41415-021-3775-4.
Rajeev, G., Lewis, Aj.N., S., 2020. A time based objective evaluation of the erosive effects of various beverages on enamel and cementum of deciduous and permanent teeth. J. Clin. Exp. Dent. e1–e8. https://doi.org/10.4317/jced.55910.
Rathee, M., Sapra, A., 2025. Dental Caries. StatPearls Publishing.
Reddy, A., Norris, D.F., Momeni, S.S., Waldo, B., Ruby, J.D., 2016. The pH of beverages in the United States. J. Am. Dent. Assoc. 147, 255–263. https://doi.org/10.1016/j.adaj.2015.10.019.
Sakae, L.O., Bezerra, S.J.C., João-Souza, S.H., Borges, A.B., Aoki, I.V., Aranha, A.C.C., Scaramucci, T., 2018. An in vitro study on the influence of viscosity and frequency of application of fluoride/tin solutions on the progression of erosion of bovine enamel. Arch. Oral. Biol. 89, 26–30. https://doi.org/10.1016/j.archoralbio.2018.01.017.
Science Buddies, De Brabandere, S., 2019. Make Boba for Bubble Tea. A Teatime Activity from Science Buddies. URL ⟨https://www.scientificamerican.com/article/make-boba-for-bubble-tea/⟩ (accessed 8.25.25).
Siraj, E.S., Pushpanjali, K., Manoranjitha, B.S., 2019. Efficacy of stevioside sweetener on pH of plaque among young adults. Dent. Res J. 16, 104–109.
Sritharan, N., Sahari, S., Sharon, C.C.S., Syubaili, M.A., 2024. Sugar taxation policy and sugar consumption in Malaysia. Food Res 8, 99–106. https://doi.org/10.26656/fr.2017.8(5).539.
Valenzuela, M.J., Waterhouse, B., Aggarwal, V.R., Bloor, K., Doran, T., 2021. Effect of sugar-sweetened beverages on oral health: a systematic review and meta-analysis. Eur. J. Public Health 31, 122–129. https://doi.org/10.1093/eurpub/ckaa147.
Várady, M., Tauchen, J., Klouček, P., Popelka, P., 2022. Effects of total dissolved solids, extraction yield, grinding, and method of preparation on antioxidant activity in fermented specialty coffee. Fermentation 8, 375. https://doi.org/10.3390/fermentation8080375.
World Health Organization (WHO), 2015. Guideline: Sugars intake for adults and children.
Zare, M., Zeinalabedini, M., Ebrahimpour-Koujan, S., Bellissimo, N., Azadbakht, L., 2024. Effect of stevia on blood glucose and HbA1C: A meta-analysis. Diabetes Metab. Syndr. Clin. Res. Rev. 18, 103092. https://doi.org/10.1016/j.dsx.2024.103092.
Zeng, L., Zhou, X., Su, X., Yang, Z., 2020. Chinese oolong tea: An aromatic beverage produced under multiple stresses. Trends Food Sci. Technol. 106, 242–253. https://doi.org/10.1016/j.tifs.2020.10.001.
Zubairi, S.I., Kasim, Z.M., Muhamad Zahiri, M.H.A., Faujan, N.H.-, 2023. Health concern of phenomenally premium boba milk tea and pearls beverages in Kuala Lumpur: sugary content and physico-chemical profiles analysis. Curr. Res Nutr. Food Sci. 11, 696–707. https://doi.org/10.12944/CRNFSJ.11.2.20.